Introduction

Digital-to-analog converters (DACs) translate binary control words into analog voltages used in mixed-signal interfaces, signal generation, calibration and embedded electronics. This project implements a compact two-bit R-2R resistor ladder followed by a custom CMOS op-amp buffer. Cadence Virtuoso and Spectre are used to examine the transfer curve, transient switching, gain, linearity and available power measurements at a nominal 1.8 V supply.

1. Circuit schematic

The original four-bit cell was adapted to two active inputs; its Cadence cell name remains DAC_4bit.

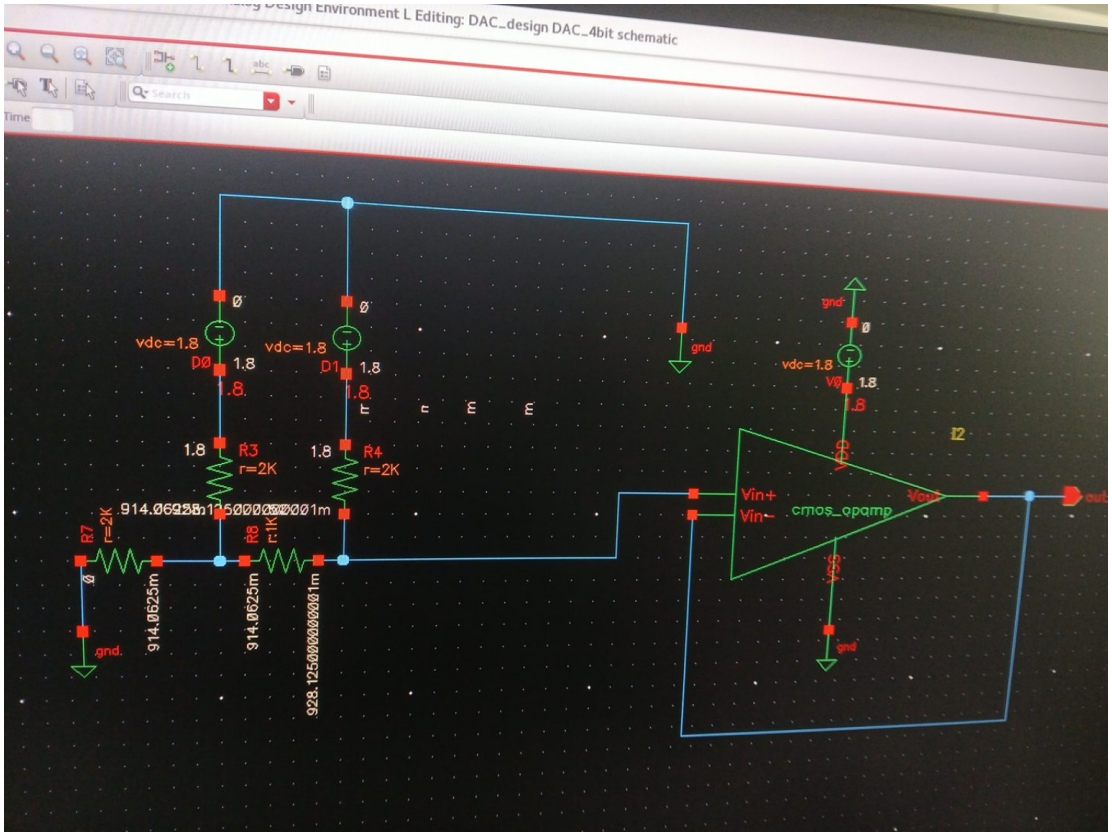


Figure 1. Final two-input R-2R DAC schematic from the supplied Cadence capture.

The ladder uses 1 kOhm and 2 kOhm nominal resistors and is buffered by a CMOS op-amp.
D1 is treated as the most significant bit; D0 is the least significant bit.
The analog output is labelled out_dac and the nominal supply is 1.8 V.

2. Design methodology

2.1 Architecture selection

An R-2R ladder is a convenient DAC architecture because it requires only two nominal resistor values, R and 2R, rather than a distinct binary-weighted resistor for each bit. In the present design, R = 1 kOhm and 2R = 2 kOhm. Each input is driven to either 0 V (logic 0) or 1.8 V (logic 1). The ladder combines these contributions into a code-dependent analog node voltage.

2.2 Buffer amplifier

The custom CMOS operational amplifier uses MOS transistors and a resistive bias arrangement. The amplifier is connected as a voltage follower in the top-level schematic, providing isolation between the resistor ladder and the output node. Its finite output swing, bias conditions, input offset, and loading can cause the actual transfer curve to differ from an ideal DAC.

2.3 Digital-to-analog conversion

For an ideal unipolar two-bit DAC with a 1.8 V reference, the output at decimal code k is $V_{ideal}(k) = k \times (1.8/4)$, for k = 0, 1, 2, 3. Thus the ideal code step (1 LSB) is **0.45 V**, and the highest ideal code produces 1.35 V. This ideal reference is used for gain and linearity comparisons; it is not a claim that the fabricated or simulated buffer can achieve these levels exactly.

Input (D1 D0)	Decimal code	Ideal voltage
00	0	0.000 V
01	1	0.450 V
10	2	0.900 V
11	3	1.350 V

2.4 Implementation sequence

The circuit was entered in Virtuoso, checked and saved, netlisted with Spectre, and simulated using DC operating-point, DC sweep, and transient analyses. The nominal 45 nm Spectre model file was used for the CMOS devices. DC outputs were recorded at the four logic combinations, and a pulsed input was used to observe switching behavior.

2.5 Design choices and practical considerations

The ladder is constructed from repeated R and 2R elements so that nominal component values remain manageable. This simplifies schematic entry and makes extension to additional bits straightforward. In a physical implementation, however, relative resistor matching is more important than absolute resistance: deviations from the intended ratio alter the weighting of individual bits and contribute to nonlinearity.

The buffer is operated as a voltage follower to reduce the loading of the ladder by the external output connection. Its real input and output characteristics still matter. A finite input common-mode range, limited output swing, bias current, and output resistance can change the measured zero-code offset and full-scale span. The reported transfer data therefore characterize the complete ladder-plus-buffer circuit, not an ideal passive ladder in isolation.

2.6 Verification strategy

The four static digital codes verify DC conversion and monotonicity; a pulsed-input transient test checks switching behavior. Comparing both with an ideal 0.45 V/LSB reference separates output offset, endpoint gain error and code-to-code step errors.

3. Simulation setup and recorded results

3.1 Simulation environment

Item	Setting
Design environment	Cadence Virtuoso ADE L with Spectre
Model library	45 nm GPDK Spectre models
Supply and logic-high level	1.8 V
Logic-low level	0 V
DC analysis	Operating point and D0 sweep (0 to 1.8 V)
Second DC sweep	D1 held at 1.8 V; D0 swept
Transient analysis	Input pulse; recorded output waveform

3.2 Measured transfer table

Code (D1 D0)	Ideal (V)	Simulated (V)	Error (V)
00	0.000	0.145538	+0.145538
01	0.450	0.497690	+0.047690
10	0.900	0.908755	+0.008755
11	1.350	1.295494	-0.054506

The two separate DC sweeps show an increasing output with D0, first with D1 low and then with D1 high. The measured values are monotonic in the verified D1 D0 ordering. The residual nonzero output at code 00 is a measured zero-code offset.

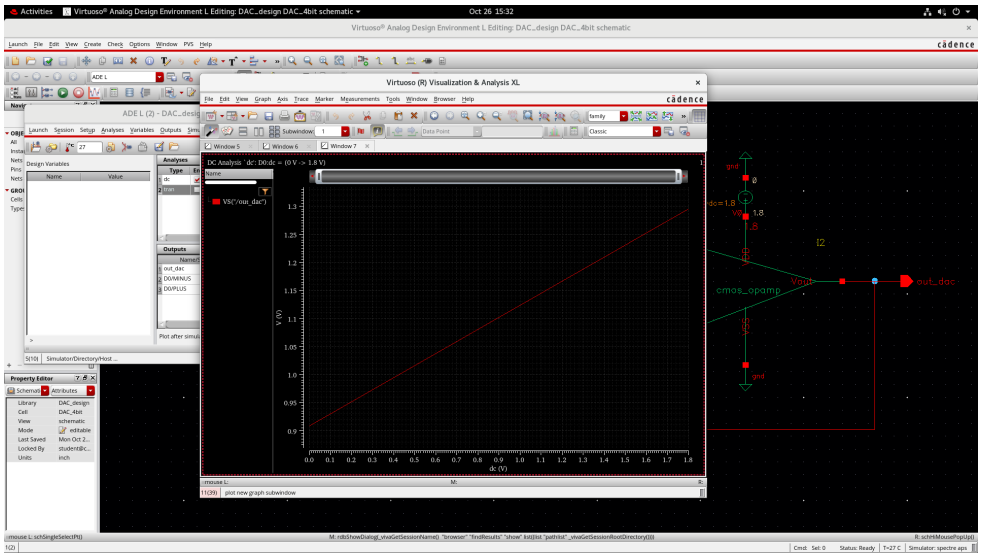


Figure 2. Cadence DC sweep with D1 high: output increases as D0 is swept (user screenshot).

4. Performance analysis: DC transfer and gain

4.1 Transfer characteristic

The transfer characteristic is the DAC output voltage versus input code. The measured curve is monotonic but differs from the ideal straight line, particularly at the lowest code. This behavior is consistent with offset and reduced output span; the data alone cannot identify the exact transistor-level cause.

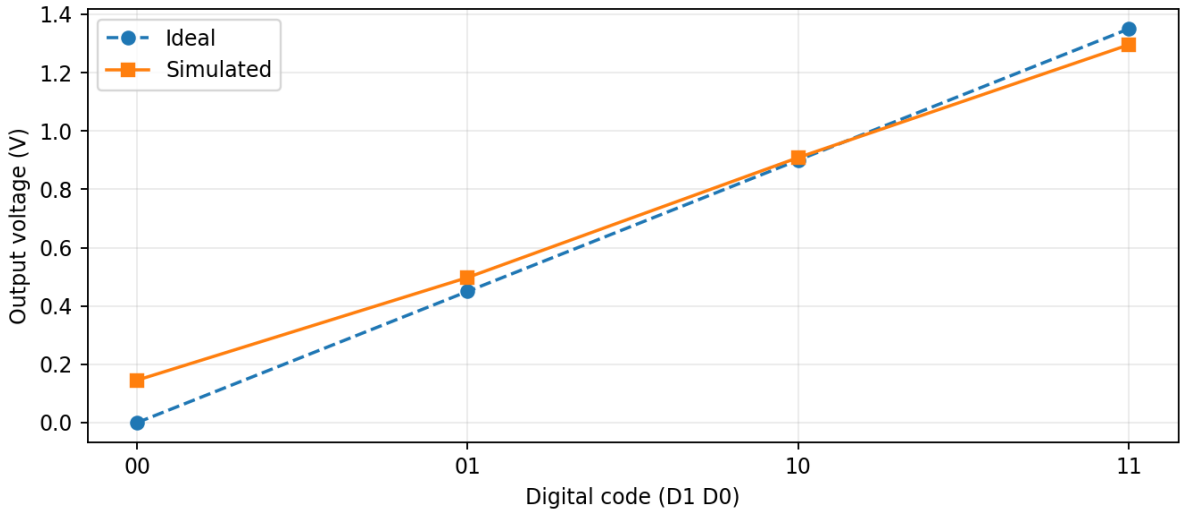


Figure 3. Ideal and simulated two-bit DAC transfer characteristics.

4.2 Effective gain and gain error

The ideal gain per code is 1.8/4 = **0.4500 V/code**. Using the measured endpoint span, the effective endpoint gain is (1.295494 - 0.145538)/3 = **0.383319 V/code**. Therefore, the gain error relative to the ideal code step is [(0.383319/0.4500) - 1] x 100 = **-14.82%**.

The zero-code offset is +0.145538 V. Because the endpoint gain calculation removes the zero-code offset by differencing, the reported gain error and offset describe separate aspects of the transfer curve. The full-scale measured output at code 11 is 1.295494 V, compared with the ideal 1.350 V.

4.3 Code-step and endpoint interpretation

Measured adjacent output increments are 0.352152 V (00 to 01), 0.411065 V (01 to 10), and 0.386739 V (10 to 11). All increments are positive, confirming monotonic behavior for the tested code sequence. They are nevertheless smaller than the ideal 0.450 V step, which is reflected in the negative differential nonlinearity values.

The endpoint output span is 1.149956 V, compared with an ideal span of 1.350 V across three code intervals. Endpoint gain therefore describes the average voltage increment per digital code. It does not imply that every individual code transition has the same slope; that distinction motivates the separate DNL analysis on the following page.

4.4 Interpreting the gain and offset

At code 00 the measured output is 0.145538 V instead of zero, indicating a positive offset. At code 11 the output is 1.295494 V, 54.506 mV below the ideal value. These deviations can coexist: offset shifts the entire transfer characteristic, while gain error changes its overall slope. Potential contributors include amplifier limitations and loading, but identifying the cause requires further circuit-level tests.

5. Performance analysis: linearity

5.1 Integral nonlinearity (ideal-reference error)

The ideal-reference INL used here is [$V_{measured}$(k) - 0.45k]/0.45, in LSB. This definition includes offset and gain errors; it is distinct from endpoint-corrected INL. The maximum absolute value is **0.323 LSB**.

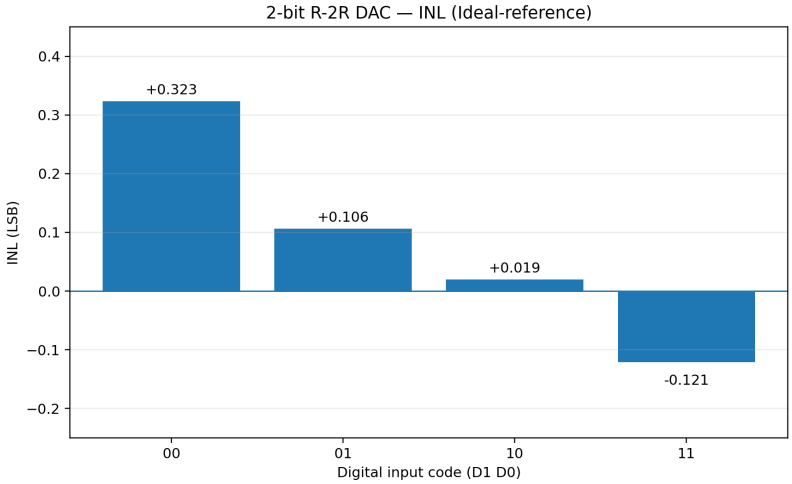


Figure 4. Ideal-reference INL calculated from the four measured DC output values.

5.2 Differential nonlinearity

DNL for transition k-1 to k is [$V_{measured}$(k) - $V_{measured}$(k-1)]/0.45 - 1, in LSB. The maximum absolute DNL is **0.217 LSB**. All three measured steps are positive, so the observed code sequence is monotonic.

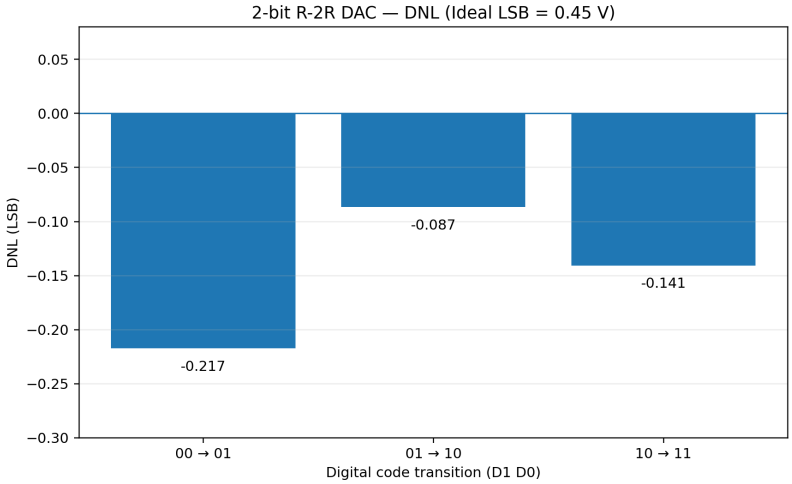


Figure 5. DNL for the three code transitions, referenced to the ideal 0.45 V LSB.

6. Performance analysis: transient response and power

6.1 Transient switching and delay

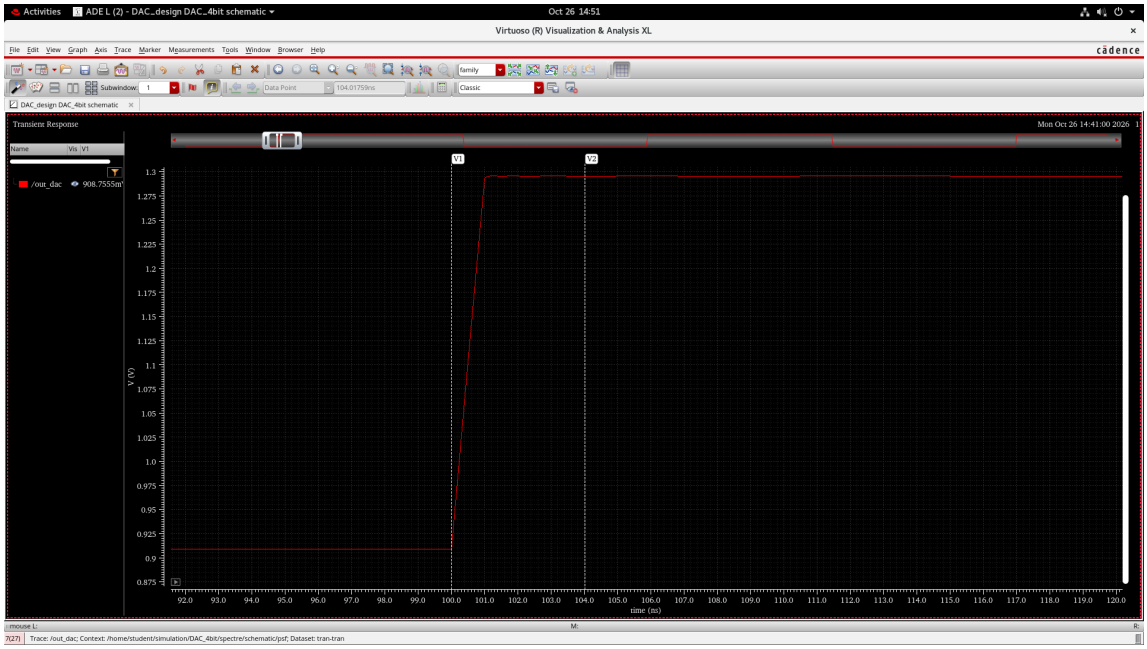


Figure 6. Cadence transient response around the 100 ns input transition (user screenshot).

The output rises from approximately 0.909 V to approximately 1.295 V after the digital input transition at 100 ns. From the screenshot, the main transition is complete near 101 ns, suggesting an **approximately 1 ns visual settling interval**. This is not a formally measured settling time: a defined tolerance band and exact waveform cursor measurements are required for a defensible delay specification.

6.2 Recorded source-power measurements

Cadence operating-point results previously captured for the source V0 include two readable readings. Power delivered by a positive supply is reported as the magnitude of the source power (Spectre may show a negative sign for a delivering source).

Readable operating-point result	Source current	Power delivered
V0, recorded reading A	-337.50 uA	607.50 uW
V0, recorded reading B	-127.406 uA	229.33 uW

Measurement limitation: *The available screenshots do not unambiguously associate these V0 readings with individual digital codes, and some Spectre log currents refer to input sources (for example D0), not the amplifier supply. Therefore a four-code total DAC power versus code plot cannot be honestly reconstructed from the verified information. It should be added only after the four V0 supply-power readings are labeled by code and any additional input-source consumption is accounted for.*

7. Comparison with existing DAC architectures

Characteristic	Implemented R-2R DAC	Resistor-string DAC	Binary-weighted resistor DAC
Resistor selection	Two nominal values: R, 2R	Many equal resistors	Several binary-weighted values
Scaling to more bits	Moderate; ladder extends	2^N elements grow rapidly	Large resistance spread
Linearity sensitivities	R/2R matching and buffer	Resistor matching and switch effects	Resistor ratio and switch effects
Practical strengths	Compact, regular topology	Inherently monotonic ideal tap selection	Conceptually simple
Practical limitations	Buffer errors; resistor mismatch	Area and switch count at higher N	Wide resistance ratios

7.1 Interpretation of measured performance

The demonstrated two-bit implementation provides four distinguishable and monotonic output levels. Its observed maximum absolute ideal-reference INL is 0.323 LSB and maximum absolute DNL is 0.217 LSB. Its endpoint gain is 0.3833 V/code, 14.82% below the ideal 0.45 V/code. These values characterize this particular simulation setup, not a generic 45 nm R-2R DAC.

7.2 Improvements and future work

Future iterations could improve output accuracy by refining the CMOS op-amp bias and output swing, matching ladder resistor ratios, and verifying switch/source impedances. A nested DC or parametric sweep can capture all four code states in one repeatable experiment. More precise delay analysis should use a defined 1% or 0.5-LSB settling band; total power analysis should include all relevant supply and digital-driver sources.

8. Conclusion

A two-bit CMOS-buffered R-2R DAC was designed and evaluated in Cadence Virtuoso. DC simulations confirmed four monotonic code outputs between 0.145538 V and 1.295494 V. The effective gain, INL, DNL, and transient response were evaluated from the supplied Spectre results. The design demonstrates successful low-resolution digital-to-analog conversion, while its offset and gain error indicate room for improvement. The power values presented are verified source readings, not a completed four-code total-power characterization.

References and data provenance

Cadence Virtuoso ADE L / Spectre simulation screenshots supplied by the student; final DAC schematic and output waveforms supplied in the conversation. Ideal R-2R DAC equations and standard definitions of gain, INL and DNL are used for analysis. No fabricated comparison measurements are included.